ShopInsight AI

E-Commerce Purchase Intelligence & Decision-Support System

CMPT 310 - Artificial Intelligence | Milestone 1 Progress Report

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1. Project Summary

ShopInsight AI is an e-commerce purchase prediction system that uses supervised binary classification to predict whether an online shopping session will result in a purchase. Using 17 browsing and visitor-related features, the system predicts **Purchase or No Purchase** and estimates the probability of a purchase. We implemented and compared Logistic Regression, Decision Tree, and K-Nearest Neighbors (KNN), and conducted additional experiments to improve the detection of less frequent purchasing sessions.

2. Work Completed & Evidence

Dataset and Preprocessing: We used the UCI Online Shoppers Purchasing Intention Dataset (12,330 sessions). After removing 125 duplicates and confirming no missing values, we retained 12,205 sessions. We created stratified training, validation, and test sets (8,543 / 1,831 / 1,831) and applied one-hot encoding and numerical scaling, fitting preprocessing only on training data.

Baseline Models: We trained Logistic Regression, Decision Tree, and KNN, evaluating them using accuracy, precision, recall, F1-score, and ROC-AUC. The Decision Tree performed best, achieving an F1-score of 0.6492 and ROC-AUC of 0.9231, but missed 113 of 286 actual purchases.

Validation baseline	Accuracy	Precision	Recall	F1 / ROC-AUC
Logistic Regression	0.8886	0.7971	0.3846	0.5189 / 0.8921
Decision Tree	**0.8979**	**0.7004**	**0.6049**	**0.6492 / 0.9231**
KNN	0.8837	0.7325	0.4021	0.5192 / 0.8067

Additional Experiments: Removing PageValues reduced the Decision Tree's F1-score from 0.6492 to 0.1173, revealing strong dependence on this feature. Balanced class weights improved purchase recall at the cost of precision. In our threshold-tuning experiment, the Decision Tree achieved a validation F1-score of 0.6799 at a threshold of 0.65. We also generated confusion matrices, ROC and precision–recall curves, and sample purchase-probability predictions.

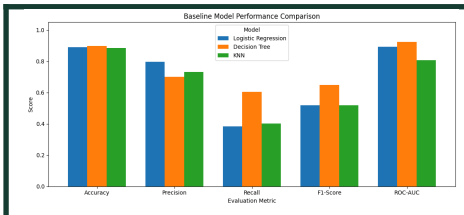


Figure 1. Baseline model comparison (validation).

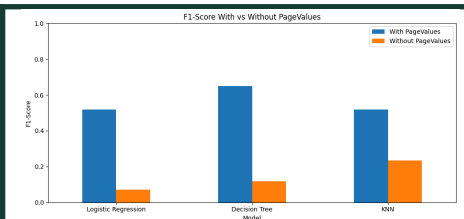


Figure 2. PageValues ablation (validation F1).

Project evidence: GitHub repository | Colab notebook | Figures & CSV results | UCI dataset

3. Challenges / Work Behind Schedule

All core Milestone 1 deliverables were completed according to our proposed scope. Our main challenges were the **class imbalance** (only ~15% of sessions resulted in purchases), which affected purchase recall, and the models' strong dependence on **PageValues**. We investigated these issues through class weighting, decision threshold tuning, and feature ablation. However, whether PageValues is available early enough for real-time prediction remains unresolved and will be investigated further in Milestone 2.

4. Changes to the Project Plan

We are retaining our original dataset and binary classification objective. Based on the professor's feedback, we expanded Milestone 1 to include **class imbalance handling, decision threshold tuning, and PageValues ablation**. For Milestone 2, we plan to implement a neural network from scratch in NumPy (forward propagation, backpropagation, and SGD), compare it with a library implementation, introduce gradient boosting, and evaluate generalization using later-month data. The held-out test set will remain reserved for final evaluation.